NXT Studio

IoT Data Analytics Internship

Week 2 Assignment

Submitted By

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Course: B.Tech – CSE (INTERNET OF THINGS)

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Submitted To

NXT Studio

Internship Duration

12 Weeks

Assignment

Week 2 - Python for Data Analytics

Table of Contents:

Program 1: Variables and Data Types

Aim:

To understand variables and different data types in Python.

Theory:

Variables are used to store data in Python. Common data types include int, float, string, and boolean.

Program:

```
[1]
0s
name = "Saniya"
age = 20
height = 5.4
student = True

print(name)
print(age)
print(height)
print(student)
```

Output:

```
Saniya
20
5.4
True
```

Conclusion: The program successfully demonstrates the use of variables and different data types in Python.

Program 2: Loops

Aim:

To understand the use of loops in python

Theory:

A loop is used to execute a block of code repeatedly. The for loop iterates over a sequence of values.

Program:

```
[3]
0s
for i in range(5):
    print(i)
```

Output:

```
0
1
2
3
4
```

Conclusion: The program demonstrates the working of a for loop by printing numbers from 0 to 4.

Program 3: Functions

Aim:

To understand the use of functions in Python.

Theory:

A function is a reusable block of code that performs a specific task. Functions help make programs more organized and reduce code repetition.

Program:

```
[4]
✓ 0s
def greet(name):
    print("Hello", name)

greet("Saniya")
```

Output:

```
Hello Saniya
```

Conclusion: The program demonstrates how to create and call a function in Python.

Program 4: File Handling

Aim:

To understand file handling in Python.

Theory:

File handling allows a program to create, read, write, and modify files. Python provides built-in functions such as open(), write(), and read() for file operations.

Program:

```
file = open("sample.txt", "r")

print(file.read())

file.close()
```

Output:

```
Hello World
```

Conclusion: The program demonstrates how to write data to a file and read it using Python.

Program 5: Exception Handling

Aim:

To understand exception handling in Python.

Theory:

Exception handling is used to handle runtime errors without stopping the execution of the program. Python uses try and except blocks to catch and handle exceptions.

Program:

```
try:
    number = 10 / 0
except ZeroDivisionError:
    print("Error: Cannot divide by zero.")
```

Output:

```
Error: Cannot divide by zero.
```

Conclusion: The program successfully demonstrates exception handling by catching a division-by-zero error.

Program 6: Introduction to NumPy

Aim:

To understand the basics of the NumPy library and perform simple array operations.

Theory:

NumPy is a Python library used for numerical computing. It provides support for arrays and mathematical operations, making data processing faster and more efficient.

Program:

```
[12]
0s

import numpy as np

marks = np.array([90, 95, 96, 88, 92])

print("Marks:", marks)
print("Highest Marks:", np.max(marks))
print("Lowest Marks:", np.min(marks))
print("Average Marks:", np.mean(marks))
print("Total Marks:", np.sum(marks))
```

Output:

```
Marks: [90 95 96 88 92]
Highest Marks: 96
Lowest Marks: 88
Average Marks: 92.2
Total Marks: 461
```

Conclusion: The program demonstrates how to create a NumPy array and calculate its average using the mean() function.

Conclusion:

During Week 2: Python for Data Analytics, I learned the fundamentals of Python programming, including variables and data types, loops, functions, file handling, exception handling, and the basics of the NumPy library. I successfully implemented and executed each program and understood their practical applications in data analytics.

References

Python Official Documentation – https://docs.python.org/3/

NumPy Official Documentation –

https://numpy.org/doc/